

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Fifth Semester of B. Tech. Examination (CL)

November 2012

CL305 Environmental Engineering-II

Date: 30.11.2012, Friday

Time: 10:00 a.m. To 01:00 p.m.

Maximum Marks: 70

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION - I

- Q-1 (a) What are different levels of treatment given to domestic sewage in a conventional sewage treatment plant? Which types of impurities are removed at each level? [03]
- (b) What are the effects of following compounds/parameters on conveyance/treatment/disposal of domestic wastewater? [04]
1. Surfactant, 2. Sulphates, 3. Biodegradable organic matter, 4. Temperature
- Q-2 (a) Draw a neat graph of BOD v/s incubation period. Derive a first-order equation for BOD remaining at any time t. sewage [05]
- (b) Estimate the average and maximum hourly flow for a community of 15,000 persons. Assume necessary data. [04]
- (c) Write short note (any one) [05]
1. Crown corrosion, 2. Self-cleansing & Limiting velocities in sewers, 3. Aerated grit chamber

OR

- Q-2 (a) Discuss various factors affecting concentration of dissolved oxygen (DO) in a natural water body. [05]
- (b) Gujarat Pollution Control Board (GPCB) received a complaint that someone has illegally discharged industrial wastewater in a dry open pit. When visiting the site, officers of GPCB were informed that the wastewater was discharged two days ago. The officers collected a sample from the pit whose 5-day BOD was 876 mg/L and reaction rate constant was 0.12 day^{-1} . The table below shows ultimate carbonaceous BOD (L_0) of 4 industries in that region. Which industry is most likely the source of wastewater discharged in the pit? Support your answer with calculations. [05]

Industry	Dairy	Fruit processing	Sugar	Meat processing
Ult. BOD (L_0)	678	1859	876	2105

- (c) Briefly describe various limitations of BOD test. [04]
- Q-3 (a) Explain with a neat sketch biological process occurring in a trickling filter. [06]
- (b) Attempt any two of the following examples. [08]
- (1) A circular sewer 50 cm in diameter is laid at a gradient of 1 in 200. Using $N=0.0125$, calculate velocity and discharge if the sewer is running full.
- (2) Design a circular/rectangular primary sedimentation tank for a sewage flow of 5.4 MLD. The detention period is 2.5 h and surface overflow rate is $40 \text{ m}^3/\text{m}^2\text{-d}$.

(3) Determine size of a trickling filter for following data:

Flow= 4 MLD, recirculation ratio= 1.4, BOD of sewage at inlet to trickling filter= 190 mg/L, treated effluent BOD desired= 30 mg/L. Assume depth of media = 1.8 m

OR

- Q-3 (a) (1) Explain with neat sketch the eco-system of a facultative pond. [03]
 (2) Differentiate between attached and suspended growth processes [03]
 (b) Attempt any one of the following examples. [05]
 (1) Design a conventional activated sludge plant to treat 30 MLD of primary settled sewage of BOD=265 mg/L. The effluent BOD required is 30 mg/L. Use following details: F/M ratio = 0.3-0.35 day⁻¹, MLSS = 2500-3500 mg/L, SVI = 95-115 mL/g.
 (2) Design a grit chamber having rectangular cross section for following data:
 (i) max. sewage flow = 20 MLD, (ii) diameter of smallest grit particle to be removed = 0.2 mm, (iii) average temperature = 20 °C, (iv) Sp. Gr. of grit particles = 2.65.
 (c) Explain mechanism of killing of pathogenic microbes in facultative ponds. [03]

SECTION - II

- Q-4 (a) Explain various kinds of injury to plants due to air pollutants. [04]
 (b) Differentiate 'Garbage' & 'Rubbish'. [03]
 Q-5 (a) Explain the term 'composting'. Give the different types of composting in use. [03]
 (b) Explain with neat sketch the mechanism of action of air pollutants on human beings. [04]
 (c) Describe settling chamber with neat sketch. [07]

OR

- Q-5 (a) Describe with neat sketch the principle, construction & working of a cyclone. [07]
 (b) Distinguish between Dry & Wet Precipitators. [03]
 (c) Explain the impacts of improper disposal of solid wastes on human health & environment in general. [04]
 Q-6 (a) Describe the effects of Carbon Monoxide & Sulfur Dioxide on human health [04]
 (b) What are the general applications of the following air pollution control equipments? [03]
 (i) Cyclones (ii) Fabric Filters (iii) Electrostatic precipitators
 (c) What is smog? What are the causes & effects of Smog on human health? [04]
 (d) What are the factors to be considered in the design of incinerators? What are the limitations of this process? [03]

OR

- Q-6 (a) Classify air pollutants into different categories, indicating their sources [05]
 (b) Define & explain the term 'Thermal pyrolysis', & its use. [05]
 (c) Write Short note on importance of activated carbon as an adsorbent. [04]

$$E = \frac{100}{1 + 0.4432 \sqrt{\frac{W1}{VF}}}$$

$$F = \frac{1 + r}{(1 + 0.1r)^2}$$

$$v = \frac{1}{n} r^{\frac{2}{3}} S^{\frac{1}{2}}$$

$$V_s = \sqrt{\frac{8\beta g}{f}} (Ss - 1)d$$

$$V_s = 60.6(Ss-1)d(3t+70)/100$$

$$H_L = 0.0729(V^2 - v^2)$$

$$E/M = S_o/X_v.t$$

$$SOR = \text{Flow/surface area}$$

$$Y_t = L_o(1 - 10^{-kt})$$

$$k_T = k_{20}(1.047)^{(T-20)}$$

$$BOD = (D_i - D_o)/P$$

$$L = L_o.10^{-kt}$$